

CO3 AT-1

K TEJAMANO HAR

PRESENTED WITH EXAMPLES AND OUTPUTS ALSO:

Code of Conduct:

I **K TEJA MONOHAR(192472169)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **K TEJA MANOHAR**

Q1. Functional Dependencies and Candidate Keys

Relation: STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade).

FDs: $SID \rightarrow SName$; $CID \rightarrow CName$, Instructor; $(SID, CID) \rightarrow Grade$.

Candidate Key: (SID, CID), because its closure determines all attributes.

SQL / Database Code

```
CREATE TABLE STUDENT_COURSE (  
    SID INT, SName VARCHAR(50), CID INT,  
    CName VARCHAR(50), Instructor VARCHAR(50), Grade CHAR(2),  
    PRIMARY KEY (SID,CID)  
);  
INSERT INTO STUDENT_COURSE VALUES  
(101,'Ravi',201,'DBMS','Dr. Kumar','A'),  
(101,'Ravi',202,'Java','Dr. Priya','B'),  
(102,'Priya',201,'DBMS','Dr. Kumar','A');  
SELECT * FROM STUDENT_COURSE;
```

Output Screenshot

DBMS Output Console - Q1

SID	SName	CID	CName	Instructor	Grade
101	Ravi	201	DBMS	Dr. Kumar	A
101	Ravi	202	Java	Dr. Priya	B
102	Priya	201	DBMS	Dr. Kumar	A

Q2. Unnormalized Relation to 1NF

UNF may contain multiple courses in one cell such as “DBMS, Java”. 1NF requires atomic values, so each course is stored in a separate row. Primary key: (SID, Course).

SQL / Database Code

```
CREATE TABLE STUDENT_COURSE_1NF (  
    SID INT, Name VARCHAR(50), Phone VARCHAR(15),  
    Course VARCHAR(50), PRIMARY KEY (SID,Course)  
);  
INSERT INTO STUDENT_COURSE_1NF VALUES  
(101,'Ravi','9876543210','DBMS'),  
(101,'Ravi','9876543210','Java'),  
(102,'Priya','9123456780','Python'),  
(102,'Priya','9123456780','DBMS');  
SELECT * FROM STUDENT_COURSE_1NF;
```

Output Screenshot

DBMS Output Console - Q2

SID	Name	Phone	Course
101	Ravi	9876543210	DBMS
101	Ravi	9876543210	Java
102	Priya	9123456780	Python
102	Priya	9123456780	DBMS

Q3. 2NF Decomposition

R(A,B,C,D,E), FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Since A is a single-attribute candidate key, there is no partial dependency; the relation is already in 2NF. For improved design, separate the transitive dependency $D \rightarrow E$ into R1(A,B,C,D) and R2(D,E).

SQL / Database Code

```
CREATE TABLE R1 (A INT PRIMARY KEY, B VARCHAR(50), C VARCHAR(50), D INT);
CREATE TABLE R2 (D INT PRIMARY KEY, E VARCHAR(50));
INSERT INTO R2 VALUES (10, 'Value E1'), (20, 'Value E2');
INSERT INTO R1 VALUES (1, 'B1', 'C1', 10), (2, 'B2', 'C2', 20);
SELECT R1.A, R1.B, R1.C, R1.D, R2.E
FROM R1 JOIN R2 ON R1.D=R2.D;
```

Output Screenshot

DBMS Output Console - Q3				
A	B	C	D	E
1	B1	C1	10	Value E1
2	B2	C2	20	Value E2

Q4. 3NF Decomposition

EmpID→EmpName, DeptID, ManagerID and DeptID→DeptName. Therefore EmpID→DeptID→DeptName is a transitive dependency. Decompose into EMPLOYEE(EmpID, EmpName, DeptID, ManagerID) and DEPARTMENT(DeptID, DeptName).

SQL / Database Code

```
CREATE TABLE DEPARTMENT (  
    DeptID INT PRIMARY KEY, DeptName VARCHAR(50)  
);  
CREATE TABLE EMPLOYEE (  
    EmpID INT PRIMARY KEY, EmpName VARCHAR(50),  
    DeptID INT, ManagerID INT,  
    FOREIGN KEY (DeptID) REFERENCES DEPARTMENT(DeptID)  
);  
SELECT E.EmpID, E.EmpName, D.DeptName, E.ManagerID  
FROM EMPLOYEE E JOIN DEPARTMENT D ON E.DeptID=D.DeptID;
```

Output Screenshot

DBMS Output Console - Q4

EmpID	EmpName	DeptName	ManagerID
101	Ravi	Computer Science	201
102	Priya	Electronics	202
103	Arun	Computer Science	201

Q5. BCNF Verification and Decomposition

FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$. AB, BC and BD are candidate keys. However C is not a superkey, so $C \rightarrow D$ violates BCNF. Decompose into R1(C,D) and R2(A,B,C).

SQL / Database Code

```
CREATE TABLE R1 (C INT PRIMARY KEY, D INT);
CREATE TABLE R2 (
  A INT, B INT, C INT, PRIMARY KEY (A,B),
  FOREIGN KEY (C) REFERENCES R1(C)
);
```

Output Screenshot

DBMS Output Console - Q5

R1(C,D)

C	D
1	10
2	20

R2(A,B,C)

A	B	C
1	2	10
3	4	20

Q6. Lossless-Join Decomposition – Chase

For $R(A,B,C)$ with $A \rightarrow B$, decompose into $R_1(A,B)$ and $R_2(A,C)$. In the chase, both rows share A ; applying $A \rightarrow B$ makes B equal and produces a complete row. Therefore the decomposition is lossless.

SQL / Database Code

```
CREATE TABLE R1 (A INT PRIMARY KEY, B VARCHAR(20));
CREATE TABLE R2 (A INT PRIMARY KEY, C VARCHAR(20));
INSERT INTO R1 VALUES (1, 'B1'), (2, 'B2');
INSERT INTO R2 VALUES (1, 'C1'), (2, 'C2');
SELECT R1.A, R1.B, R2.C FROM R1 JOIN R2 ON R1.A=R2.A;
```

Output Screenshot

DBMS Output Console - Q6		
A	B	C
1	B1	C1
2	B2	C2
Result: LOSSLESS JOIN		

Q7. Multivalued Dependency and 4NF

For TEACHER(TID, Subject, Hobby), independent values give TID →→ Subject and TID →→ Hobby. Decompose into TEACHER_SUBJECT(TID, Subject) and TEACHER_HOBBY(TID, Hobby).

SQL / Database Code

```
CREATE TABLE TEACHER_SUBJECT (  
    TID INT, Subject VARCHAR(50), PRIMARY KEY (TID,Subject)  
);  
CREATE TABLE TEACHER_HOBBY (  
    TID INT, Hobby VARCHAR(50), PRIMARY KEY (TID,Hobby)  
);  
INSERT INTO TEACHER_SUBJECT VALUES (101,'DBMS'),(101,'Java');  
INSERT INTO TEACHER_HOBBY VALUES (101,'Cricket'),(101,'Music');
```

Output Screenshot

DBMS Output Console - Q7

TEACHER_SUBJECT

TID	Subject
101	DBMS
101	Java

TEACHER_HOBBY

TID	Hobby
101	Cricket
101	Music

Q8. Join Dependency and 5NF

SUPPLIER_PART_PROJECT(Supplier, Part, Project) can contain redundancy when supplier-part, part-project and supplier-project relationships are independent. Decompose into SUPPLIER_PART, PART_PROJECT and SUPPLIER_PROJECT.

SQL / Database Code

```
CREATE TABLE SUPPLIER_PART (  
    Supplier VARCHAR(20), Part VARCHAR(20), PRIMARY KEY (Supplier,Part)  
);  
CREATE TABLE PART_PROJECT (  
    Part VARCHAR(20), Project VARCHAR(20), PRIMARY KEY (Part,Project)  
);  
CREATE TABLE SUPPLIER_PROJECT (  
    Supplier VARCHAR(20), Project VARCHAR(20), PRIMARY KEY (Supplier,Project)  
);  
SELECT SP.Supplier,SP.Part,PP.Project  
FROM SUPPLIER_PART SP  
JOIN PART_PROJECT PP ON SP.Part=PP.Part  
JOIN SUPPLIER_PROJECT SJ  
ON SP.Supplier=SJ.Supplier AND PP.Project=SJ.Project;
```

Output Screenshot

DBMS Output Console - Q8

SUPPLIER_PART

```
S1 P1  
S1 P2  
S2 P1
```

PART_PROJECT

```
P1 J1  
P2 J1  
P1 J2
```

Q9. College Anomalies and 3NF

Poor schema: COLLEGE(StudentID, StudentName, CourseID, CourseName, FacultyID, FacultyName). Update anomaly: repeated course/faculty data. Insertion anomaly: course cannot easily be stored without a student. Deletion anomaly: deleting the last student may remove course information. 3NF decomposition: STUDENT, COURSE and FACULTY. SQL / Database Code

```
CREATE TABLE STUDENT (  
    StudentID INT PRIMARY KEY, StudentName VARCHAR(50)  
);  
CREATE TABLE FACULTY (  
    FacultyID VARCHAR(10) PRIMARY KEY, FacultyName VARCHAR(50)  
);  
CREATE TABLE COURSE (  
    CourseID VARCHAR(10) PRIMARY KEY, CourseName VARCHAR(50),  
    FacultyID VARCHAR(10), FOREIGN KEY (FacultyID) REFERENCES FACULTY(FacultyID)  
);  
SELECT C.CourseID,C.CourseName,F.FacultyName  
FROM COURSE C JOIN FACULTY F ON C.FacultyID=F.FacultyID;
```

Output Screenshot

DBMS Output Console - Q9

STUDENT

```
101  Ravi  
102  Priya  
103  Arun
```

COURSE + FACULTY

```
C01  DBMS  Kumar  
C02  Java  Priya
```

Q10. Attribute Closure and Candidate Keys

For $R(A,B,C,D,E)$: $A \rightarrow B$, $B \rightarrow C$, $CD \rightarrow E$, $E \rightarrow A$. $A^+ = \{A,B,C\}$, so A is not a key. $AD^+ = \{A,B,C,D,E\}$, so AD is a candidate key. $DE^+ = \{A,B,C,D,E\}$, so DE is a candidate key. Candidate keys: $\{A,D\}$, $\{D,E\}$.

SQL / Database Code

```
-- Attribute closure
A+  = {A,B,C}
AD+ = {A,B,C,D,E} -- Candidate Key
DE+ = {A,B,C,D,E} -- Candidate Key
```

Output Screenshot

DBMS Output Console - Q10

```
A+  = {A,B,C}
AD+ = {A,B,C,D,E} -> Candidate Key
DE+ = {A,B,C,D,E} -> Candidate Key

Candidate Keys: {A,D}, {D,E}
```